A REALIZATION OF F_4 FOR LITT PROBLEM NO. 13

OPENAI

Abstract. Let F be the Fano surface of a smooth cubic threefold with an Eckardt point, and let $E \subset F$ parametrize the lines through that point. The image X of F in $\mathrm{Alb}(F)/E$ is an integral surface whose full convolution Tannaka group is F_4 on its minimal 26-dimensional module. The quotient contracts precisely E and splits the Albert module as $27 = 1 \oplus 26$. The fixed vector has nonzero Albert norm, as follows from the Hessian ranks on the singular cubic and connected irreducibility. The construction gives a seven-dimensional family of polarized ambient fourfolds. Its intersection complexes have characteristic cycle $\Lambda_X + 2\Lambda_0$ and generic twisted Hodge multiplicities $(6, 14, 6)$.

1. Introduction

For an integral subvariety Z of a complex abelian variety, the intersection complex IC_Z generates a Tannakian category under convolution, after objects of Euler characteristic zero are discarded. Litt Problem No. 13 asks whether the simple types G_2, E_7, E_8, and F_4 occur as the resulting groups [10]. The construction below realizes F_4 by singular surfaces.

An Eckardt point of a smooth cubic threefold is a point whose tangent hyperplane section is a cone over a smooth plane cubic. Let $Y \subset \mathbf{P}^4$ be such a threefold, with Eckardt point p. Its Fano surface $F = F(Y)$ contains the elliptic curve E of lines through p. Choose an Albanese embedding $F \hookrightarrow A = \mathrm{Alb}(F)$ based at a point of E, so that $E \subset A$ is an abelian subvariety, and put

$$q : A \longrightarrow B = A/E, \qquad X = q(F)_{\mathrm{red}}, \qquad f = q|_F.$$

Theorem 1.1. *For every smooth cubic threefold with a marked Eckardt point, B is an abelian fourfold and $X \subset B$ is an integral symmetric surface with trivial translation stabilizer. The map f is birational, its only positive-dimensional fibre is $f^{-1}(0) = E$, and*

$$Rf_*\mathbf{C}_F[2] \simeq \mathrm{IC}_X \oplus \delta_0, \qquad \chi(\mathrm{IC}_X) = 26.$$

The full convolution Tannaka pair is

$$\big(G(\mathrm{IC}_X), \omega(\mathrm{IC}_X)\big) \simeq (F_4, V_{26}),$$

where V_{26} is the minimal representation of F_4.

The proof uses the E_6 Tannaka group of the Fano surface. The only non-formal quotient step is the restricted tensor pushforward in Lemma 2.2; its hypothesis is verified by the semismall decomposition. Quotient pushforward gives

Date: October 21, 2026.

2020 *Mathematics Subject Classification.* 14K12, 14J30, 18M25, 20G41.

Key words and phrases. Fano surface, Eckardt point, intersection complex, convolution, Tannaka group, Albert algebra.

$H = G(\mathrm{IC}_X) \subset E_6$ and a restriction of the Albert module $J|_H = 1 \oplus V_{26}$. Triviality of the translation stabilizer makes V_{26} irreducible on H^0. The Hessian of the Albert norm then forces the fixed line to contain a norm-one vector. Its full stabilizer is F_4, and the classification of irreducible subgroups of the minimal module gives $H = F_4$.

2. Convolution and the normalized Fano surface

All varieties are over $\mathbf{C}$. Intersection complexes have the perverse normalization: if $j : Z_{\mathrm{reg}} \hookrightarrow Z$ and $d = \dim Z$, then $\mathrm{IC}_Z = j_{!*}\mathbf{C}_{Z_{\mathrm{reg}}}[d]$, extended by zero to the ambient abelian variety. Write δ_0 for the skyscraper sheaf at the origin. For addition m the convolution product is $K * L = Rm_*(K \boxtimes L)$.

The quotient by negligible objects is Tannakian on every rigid subcategory generated by a semisimple perverse sheaf of geometric origin [8, 5]. Its group is reductive, and $\dim \omega(K) = \chi(K)$. Proper pushforward along a homomorphism of abelian varieties commutes with convolution, but it need not preserve the entire derived quotient by Euler-negligible complexes. The restricted form proved in Lemma 2.2, applicable to the semismall map below, supplies a tensor functor and a closed immersion

$$G(Rq_*K) \hookrightarrow G(K). \tag{2.1}$$

This distinction from the cohomological quotient in [8, Lemma 13.7] is essential. Groups refer throughout to the full group; E_6 denotes its simply connected form.

The Albanese variety of F has dimension five, and its Albanese maps are embeddings [1]. For $s \in F$ let $L_s \subset Y$ be its line. Roulleau constructs an involution $\sigma = \sigma_E$ fixing E pointwise and a map $\gamma_E : F \to E$ such that, for general $s \notin E$,

$$Y \cap \langle p, L_s \rangle = L_s + L_{\gamma_E(s)} + L_{\sigma(s)}. \tag{2.2}$$

The induced Albanese automorphism has eigenvalue $+1$ on T_0E and -1 on T_0A/T_0E [11, Section 3.1 and Theorem 13]. Thus $q(\sigma(s)) = -q(s)$ and $X = -X$.

Lemma 2.1. *The Albanese embedding can be chosen with $E \subset A$ an abelian subvariety, the sum of every coplanar triple equal to zero, and $G(\mathrm{IC}_F) = E_6$ on the Albert module J.*

Proof. For an embedding based at $e \in E$, addition is constant on the incidence variety $\mathcal{T} \subset F^3$ of ordered triples forming a plane section [1, 7]; call the value $c \in A$. By (2.2), its image in B is $x + 0 - x = 0$, so $c \in E$. Choose $u \in E$ with $3u = c$ and translate the embedding by $-u$. This preserves E and the quotient and makes the triple sum zero.

The incidence correspondence now gives a nonzero invariant symmetric cubic on J [7, Example 2.2(c)]. The derived connected Tannaka group and its representation are (E_6, J) [6, Theorem 2]. On this module the invariant cubic is a nonzero multiple of the Albert norm, whose full linear stabilizer is E_6 [7, Theorem 2.1(c)]. Faithfulness therefore gives $E_6 \subset G(\mathrm{IC}_F) \subset E_6$. □

Lemma 2.2 (Restricted quotient pushforward)**.** *Let $q : A \to B$ be a homomorphism of complex abelian varieties and let K be a semisimple geometric-origin perverse sheaf on A. Assume Rq_*K is a semisimple perverse sheaf on B. Then proper pushforward induces an exact symmetric tensor functor*

$$\langle K \rangle_{\otimes} \longrightarrow \langle Rq_*K \rangle_{\otimes}$$

on the semisimple categories of nonnegligible perverse summands, and a closed immersion $G(Rq_*K) \hookrightarrow G(K)$.

Proof. Work in the quotient derived categories modulo complexes whose perverse cohomology has Euler characteristic zero, as in [5, Proposition 3.1]. Convolution is perverse t-exact there. Proper pushforward respects convolution, the unit, duality and direct sums. Every convolution word W in $K, K^\vee$ therefore maps to a convolution word in $Rq_*K, (Rq_*K)^\vee$, which is perverse in the target quotient.

By the decomposition theorem, W is a direct sum of shifted simple perverse sheaves. The nonnegligible summands are unshifted in the source quotient. Let $N[j]$ be a negligible summand. Its pushforward is a direct summand of the perverse object Rq_*W in the target quotient, so it too is perverse there. Since

$$\chi(Rq_*N[j]) = (-1)^j\chi(N) = 0,$$

its image is zero: Euler characteristic is the dimension of a perverse object in the Tannakian quotient. Thus pushforward kills every negligible summand occurring in any generating word, and sends the remaining summands into the perverse heart. The induced operation on the Karoubian tensor-generated semisimple category respects morphisms and is exact, rigid and symmetric monoidal. Its image tensor-generates $\langle Rq_*K\rangle_\otimes$. Tannaka duality gives the claimed closed immersion. □

3. The quotient geometry

The tangent bundle theorem identifies

$$\mathbf{P}(T_sF) = L_s \subset \mathbf{P}(T_0A) = \mathbf{P}^4, \qquad \mathbf{P}(T_0E) = \{p\} \tag{3.1}$$

[1][11, Proposition 4].

Proposition 3.1. *The map* $f : F \to X$ *is birational and immersive on* $F \setminus E$. *Its only contracted curve is* E, *and* $f^{-1}(0) = E$.

Proof. By (3.1), $\ker(df_s) \neq 0$ exactly when $p \in L_s$, that is, when $s \in E$. Any contracted curve consequently lies in E. Conversely, E is contracted, and $q^{-1}(0) = E$ in A gives the stated fibre. Thus f is generically finite onto an integral surface.

Choose a dense open $U \subset X_{\mathrm{reg}}$ where f is finite and unramified, avoiding $B[2]$ and the nongeneric locus of (2.2). For $s, t \in f^{-1}(x)$ with $x \in U$, projection from p sends both L_s and L_t to $\mathbf{P}(T_xX)$. Hence both lie in $\Pi = \langle p, L_s\rangle$. The plane section (2.2) has exactly three lines. Its line through p maps to zero, so $t = s$ or $t = \sigma(s)$. The latter implies $x = -x$, contrary to $x \notin B[2]$. The generic degree is therefore one. □

Proposition 3.2. *One has* $Rf_*\mathbf{C}_F[2] \simeq \mathrm{IC}_X \oplus \delta_0$ *and* $\chi(\mathrm{IC}_X) = 26$.

Proof. The proper birational surface map f is semismall. Its only fibre of positive dimension is the irreducible curve E over zero. Roulleau computes $E^2 = -3$ [11, Proposition 10]. The local intersection form is the nondegenerate matrix (-3), so the semismall decomposition has exactly one skyscraper summand [3, Theorem 3.4.1]. Since $e(F) = 27$ [1], the Euler characteristic of the remaining summand is $27 - 1 = 26$. □

Proposition 3.3. $\mathrm{Stab}_B(X) = \{0\}$, *and the tautological module of* $G(\mathrm{IC}_X)$ *is irreducible on the identity component.*

Proof. Factor f through the normalization $\nu : X^\nu \to X$, with $h : F \to X^\nu$. The only exceptional curve of h is E. A proper birational morphism of normal surfaces is an isomorphism outside its exceptional locus, so $X^\nu \setminus \{z\} \simeq F \setminus E$, where $z = h(E)$. The point z is singular: a birational morphism to a smooth surface factors locally into point blowups, whose exceptional curves are rational, whereas E is elliptic. Thus $\mathrm{Sing}(X^\nu) = \{z\}$.

A translation preserving X lifts to its normalization and fixes z. Since $\nu(z) = 0$, the translation is zero. This is precisely nondivisibility of the simple, non-negligible sheaf IC_X. Connected irreducibility follows from [5, Proposition 3.3(2)]. □

4. The Albert norm and the full group

Put $H = G(\mathrm{IC}_X)$ and $V = \omega(\mathrm{IC}_X)$. By Lemma 2.1 and Proposition 3.2 and (2.1),

$$H \hookrightarrow E_6, \qquad J|_H \simeq 1 \oplus V, \qquad \dim V = 26. \tag{4.1}$$

Indeed the tensor unit already belongs to $\langle \mathrm{IC}_X \rangle$, so $\langle \mathrm{IC}_X \oplus \delta_0 \rangle = \langle \mathrm{IC}_X \rangle$. Let $v_0 \neq 0$ span the trivial summand, and let N be the Albert cubic. Its complete polarization defines $\beta_v : J \to J^\vee$ by $\beta_v(u) = N(v, u, -)$.

Lemma 4.1. *For $v \neq 0$ with $N(v) = 0$, the kernel of β_v has dimension 17 in Albert rank one and 9 in Albert rank two.*

Proof. In octonion coordinates the cubic is

$$N = abc - a\,n(\xi) - b\,n(\eta) - c\,n(\zeta) + 2\,\mathrm{Re}((\xi\eta)\zeta),$$

where each octonion norm is nondegenerate of rank eight. The rank-orbit description of the Albert algebra reduces the two cases to $e_1 = \mathrm{diag}(1, 0, 0)$ and $e_1 + e_2$ [13]. The quadratic terms at these points are, respectively,

$$bc - n(\xi), \qquad \gamma(\alpha + \beta) - n(\xi) - n(\eta).$$

Their ranks are 10 and 18, giving kernel dimensions $27 - 10$ and $27 - 18$. □

Proposition 4.2. *The fixed vector satisfies $N(v_0) \neq 0$.*

Proof. The map β_{v_0} is H^0-equivariant. By Proposition 3.3, $J|_{H^0} = 1 \oplus V$ is a multiplicity-free sum of two irreducibles. Reductivity therefore gives

$$\dim \ker \beta_{v_0} \in \{0, 1, 26, 27\}.$$

If $N(v_0) = 0$, the vector has Albert rank one or two, contradicting Lemma 4.1. □

Scale v_0 to norm one. Its full stabilizer in E_6 is the connected group F_4 [13]. Hence $H \subset F_4$, and V is its minimal module. The group H^0 is positive-dimensional, since it acts irreducibly on a space of dimension 26. No proper positive-dimensional closed subgroup of F_4 in characteristic zero acts irreducibly on this module: this is the F_4 row of [9, Theorem 1 and Table 1.2]. Thus

$$F_4 = H^0 \subset H \subset F_4.$$

This proves Theorem 1.1, including connectedness of the full group.

5. The family and its numerical invariants

Corollary 5.1. *The construction yields a seven-dimensional family of polarized ambient fourfolds of type* $(1,2,2,2)$. *Every member carries the surface of Theorem 1.1.*

Proof. For a smooth cubic surface $S=\{g=0\}\subset\mathbf{P}^3$ and a transverse hyperplane $\{\ell=0\}$, the cubic $Y=\{g+x_0^2\ell=0\}\subset\mathbf{P}^4$ is smooth and has Eckardt point $[1:0:0:0:0]$. Let $P=\mathrm{Im}(1-\sigma)\subset J(Y)=A$. The invariant part is E, and the principal polarization identifies A/E with $P^\vee$. The polarization on P has type $(1,1,1,2)$; its dual has type $(1,2,2,2)$ [2, Lemma 1.12 and Remark 1.14]. The moduli space of these cubic surface pairs has dimension seven, and its period map to polarized P is injective with injective differential [2, Theorem 0.1]. Dualization gives the asserted dimension for the family of B. □

Corollary 5.2 (A concrete example)**.** *The cubic* $x_0^2x_1+x_1^3+x_2^3+x_3^3+x_4^3=0$, *marked at* $p=[1:0:0:0:0]$, *gives a realization of* (F_4,V_{26}).

Proof. The partial derivatives are $2x_0x_1$, $x_0^2+3x_1^2$, and $3x_i^2$ for $i=2,3,4$; they have no common projective zero. The tangent section at p is the cone $x_2^3+x_3^3+x_4^3=0$ in $x_1=0$. Apply Theorem 1.1. □

For a subvariety $Z\subset B$, write Λ_Z for the closure of the conormal bundle to Z_{reg} in T^*B.

Proposition 5.3. *The characteristic cycle and conormal Gauss degree are*

$$\mathrm{CC}(\mathrm{IC}_X)=\Lambda_X+2\Lambda_0,\qquad \deg(\Lambda_X\to T_0^*B)=24.$$

Proof. Over $B\setminus\{0\}$, f is finite and immersive. Locally its image is a union of smooth branches, and its direct image is the sum of their constant intersection complexes. By Proposition 3.2, no other conormal component occurs away from zero. Thus $\mathrm{CC}(\mathrm{IC}_X)=\Lambda_X+m\Lambda_0$.

A general covector in $T_0^*B\subset T_0^*A$ defines a general hyperplane $H\subset\mathbf{P}^4$ through p. By (3.1), a conormal point away from zero is a line $L_s\subset Y\cap H$ not passing through p. The surface $Y\cap H$ is a smooth cubic surface with Eckardt point p. Of its 27 lines, exactly three pass through p. The remaining 24 give the generic Gauss fibre; exceptional strata are avoided by a general covector. Hence the degree is 24. The index formula on abelian varieties [4, Theorem 1.3] now gives $26=24+m$. □

Proposition 5.4. *For a general unitary character* α *of* B, *the twisted hypercohomology* $H^0(B,\mathrm{IC}_X\otimes\mathcal{L}_\alpha)$ *is pure of weight two with Hodge multiplicities* $(6,14,6)$.

Proof. Generic vanishing and Proposition 3.2 give $H^i(F,f^*\mathcal{L}_\alpha)=0$ for $i\neq 2$ and dimension 27 in degree two [8, Theorem 1.1]. The unitary Hodge decomposition therefore makes the higher cohomology of $K_F\otimes f^*\alpha$ vanish. Riemann–Roch gives $h^0(K_F\otimes f^*\alpha)=\chi(K_F)=6$, since $p_g(F)=10$ and $q(F)=5$ [1]. The same calculation for α^{-1} and duality give Hodge multiplicities $(6,15,6)$ for the twisted degree-two cohomology of F.

The rational Hodge module form of the semismall decomposition is $Rf_*\mathbf{Q}_F^H[2]=\mathrm{IC}_X^H\oplus\mathbf{Q}_0^H(-1)$ [3, 12]. The last term has type $(1,1)$, yielding $(6,14,6)$ after its removal. □

Remark 5.5. The 24 short roots and the zero weight of multiplicity two in V_{26} agree with the two terms of the characteristic cycle. The numerical calculation is independent of the subgroup argument.

References

[1] C. H. Clemens and P. A. Griffiths, *The intermediate Jacobian of the cubic threefold*, Ann. of Math. (2) **95** (1972), 281–356.

[2] S. Casalaina-Martin and Z. Zhang, *The moduli space of cubic surface pairs via the intermediate Jacobians of Eckardt cubic threefolds*, J. Lond. Math. Soc. (2) **104** (2021), no. 1, 1–34.

[3] M. A. A. de Cataldo and L. Migliorini, *The hard Lefschetz theorem and the topology of semismall maps*, Ann. Sci. École Norm. Sup. (4) **35** (2002), no. 5, 759–772. doi:10.1016/S0012-9593(02)01108-4.

[4] J. Franecki and M. Kapranov, *The Gauss map and a noncompact Riemann–Roch formula for constructible sheaves on semiabelian varieties*, Duke Math. J. **104** (2000), no. 1, 171–180.

[5] A. Javanpeykar, T. Krämer, C. Lehn and M. Maculan, *The monodromy of families of subvarieties on abelian varieties*, Duke Math. J. **174** (2025), no. 6, 1045–1149. doi:10.1215/00127094-2024-0053.

[6] T. Krämer, *Cubic threefolds, Fano surfaces and the monodromy of the Gauss map*, Manuscripta Math. **149** (2016), 303–314. doi:10.1007/s00229-015-0785-z.

[7] T. Krämer, *Characteristic cycles and the microlocal geometry of the Gauss map, I*, Ann. Sci. École Norm. Sup. (4) **55** (2022), no. 6, 1475–1527. doi:10.24033/asens.2522.

[8] T. Krämer and R. Weissauer, *Vanishing theorems for constructible sheaves on abelian varieties*, J. Algebraic Geom. **24** (2015), no. 3, 531–568. doi:10.1090/jag/645.

[9] M. W. Liebeck and G. M. Seitz, *Subgroups of exceptional algebraic groups which are irreducible on an adjoint or minimal module*, J. Group Theory **7** (2004), no. 3, 347–372. doi:10.1515/jgth.2004.012.

[10] D. Litt, *Problem No. 13*, Problems I Like, `https://www.problemsilike.com/13`, accessed September 30, 2026.

[11] X. Roulleau, *Elliptic curve configurations on Fano surfaces*, Manuscripta Math. **129** (2009), 381–399. doi:10.1007/s00229-009-0264-5.

[12] M. Saito, *Modules de Hodge polarisables*, Publ. Res. Inst. Math. Sci. **24** (1988), no. 6, 849–995.

[13] T. A. Springer and F. D. Veldkamp, *Octonions, Jordan Algebras and Exceptional Groups*, Springer Monographs in Mathematics, Springer, Berlin, 2000. doi:10.1007/978-3-662-12622-6.